

Zeyu (Alban) LI

Nucleic Acid Scientist | Formulation, Delivery & Process Development | 626-826-5327 | zl788@cornell.edu | zeyuli.net |
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Nucleic acid scientist bridging polymer formulation and DNA/RNA process development--nucleic acid encapsulation, continuous mRNA biomanufacturing, and downstream purification. Full-cycle R&D from upstream IVT production through polymeric formulation to bead-based purification, with expertise in crosslinking chemistry, analytical characterization (qPCR, rheometry, SEM), and cleanroom processing. Engineered a 100-day continuous mRNA platform producing 6 RNA species (Nature under review) and formulated DoD-funded PLGA microparticles for nucleic acid delivery (first-author, ES&T). Co-authored four publications and co-invented two patents.

SKILLS

Molecular & Bioengineering: PCR; qPCR; IVT; Enzymatic assays and workflows; Electrophoresis (PAGE, agarose); DNA/RNA extraction & purification; Nucleic acid quantification (NanoDrop, UV-Vis); LSPR-based binding assays; Cleanroom protocols; Liquid Chromatography; Fluorometry (Qubit, Plate Reader).

Materials Characterization: SEM; DLS; EDS; Optical microscopy; UV-Vis spectroscopy; Fluorescence microscopy; Oscillatory rheometry; Tensile testing (Instron); Contact-angle measurement.

Fabrication & Processing: Electrospinning; 3D printing (DIW, SLA, FDM); Thermal imprinting; Double-emulsion microencapsulation; Hydrogel formulation; UV photocrosslinking; Spin coating; Mold casting; Photolithography; Two-photon lithography; Etching; Nanopatterning; Freeze-drying; Roll-to-roll processing

Programming & Data/Modeling: Python; Data analysis & visualization (pandas, NumPy, seaborn); Machine learning (TensorFlow, PyTorch - working knowledge); Java; AI tools (computational workflows, LLM-assisted data analysis)

2D/3D Design & Visualization: CAD; SketchUp; Autodesk 3ds Max; Blender; KeyShot; Adobe Illustrator, Photoshop, InDesign; Bootstrap Studio; Scientific figure production; Video/audio editing.

EDUCATION

Cornell University, Ithaca, NY

Biological and Environmental Engineering

Concentration: *Biological Engineering, Materials Science, Bioenvironmental Engineering*

Ph.D. -- DNA Materials Lab, Advisor: Dan Luo | Expected May 2026

M.S. | Aug 2024

M.Eng. | May 2020

Hong Kong Baptist University, Kowloon, Hong Kong

Chemistry (Major) and Computer Science (Minor)

B.Sc. (Hons) -- Microfabrication & Surface Materials Lab, Supervisor: Kangning Ren | Nov 2019

EXPERIENCE

Graduate Research Assistant | Ithaca, NY

DNA Materials Lab, Cornell University | Feb 2021 - Present

- **Formulated biodegradable PLGA microparticles for nucleic acid encapsulation via double-emulsion solvent evaporation;** produced gram-scale batches and led a DoD-funded field deployment across 11 km². Achieved qPCR detection 7 km downstream from less than 1 mg DNA release with contamination-controlled processing in a positive-pressure cleanroom. (first-author, ES&T, 2025).

- **Engineered a continuous-flow mRNA biomanufacturing platform (NucleoMesh) sustaining 100-day steady-state production across 6 diverse RNA species at milligram-scale yield:** co-designed chip reactor and independently designed a microfluidic RNA logic gate. Led downstream process development: developed a DEAE-Silica bead-based RNA purification method achieving up to 91% recovery. (2nd author, Nature under review.)
- **Invented a high-throughput nucleic acid purification platform** (pod-based, perfluorocarbon medium); delivered ~200 mg DNA per pod and reduced processing cost by ~91% vs. conventional ethanol precipitation, enabling scalable, low-cost purification for downstream qPCR and sequencing workflows.
- **Developed stimuli-responsive DNA hydrogels with three crosslinking chemistries--thermal, photochemical (UV-reversible psoralen), and ionic (Al³⁺)--**spanning thermoreversible to irreversible networks from a single biocompatible biopolymer feedstock. Optimized formulation across 6 variables; characterized by oscillatory rheometry and tensile testing. (co-first-author, ACADIA, 2024).
- **Mentored 3 undergraduate researchers and co-instructed Design and Analysis of Biomaterials** (~50% of content, ~45 students): designed a 6-lecture AI tooling module and organized a 40-student Biomaterials Hackathon with oral defense format. Presented research at ACS Fall 2023 and international forums; teaching assistant across 7 engineering courses total.

Undergraduate Research Assistant & Senior Research Assistant | Kowloon, Hong Kong

Hong Kong Baptist University | Jun 2017 - Aug 2019 & Oct 2020 - Jan 2021

- **Engineered durable monolithic perfluoropolymer surfaces with hierarchical nano-micro topologies** via thermal imprinting with master molds from photolithography, two-photon lithography, and etching; validated durability across chemical, thermal, and mechanical stress testing. (3rd-author paper, The Innovation IF 33.2; US Patent 11,839,998).
- **Led an experimental study on static-charge anti-icing**, demonstrating ~25% reduction in droplet contact time. Received the Best Undergraduate Thesis Award.

Research Exchange Trainee | Atlanta, GA

Georgia State University, Molecular Basis of Disease Program | Jun 2018 - Aug 2018

- **Characterized protein-DNA binding kinetics using LSPR** (localized surface plasmon resonance) assays, a label-free real-time technique analogous to SPR. Quantified sequence-specific binding affinities of transcription factor PU.1 to DNA targets. Received the Best Poster Presentation Award.

PUBLICATIONS

- Wang, D., [Li, Z.](#), Li, J., Han, Y., Sun, T., Li, F., Liu, P. "A Chip Reactor for Perpetual Nucleic Acid Production and On-chip Information Processing." Nature, under review (2026).
- [Li, Z.](#), Ramón, C. L., Koeberle, A., et al. "Tracing Environmental DNA Transport in Large Lakes with Synthetic DNA Microparticles and Hydrodynamic Modeling" Environmental Science & Technology (IF = 12.4), (2025). [DOI](#)
- He, C. *, [Li, Z.](#) *, Wang, L. X., et al. "PolyTile 4.0: Self-healing Ceramic Tiles" ACADIA (2024). [DOI](#)
- Li, W., Chan, C. W., [Li, Z.](#), et al. "All-perfluoropolymer, nonlinear stability-assisted monolithic surface combines topology-specific superwettability with ultradurability." The Innovation (IF = 33.2), 4(2), 100299, (2023). [DOI](#)

PATENTS

- Ren, K., Wu, H., Wang, Z., Yao, S., Ong, B., Li, W., [Li, Z.](#), Sun, H., & Chan, C.W. "Crack engineering as a new route for the construction of arbitrary hierarchical architectures." US Patent 11,839,998, (2023).
- Li, Q., [Li, Z.](#), & Lin, Z. "Reactor and method of spiral propulsion biomass continuous thermal cracking." Chinese Patent 201711214139.6 (2017).